



Does conditioning leave a structural memory?

Large-area evidence for the dislocation picture

Yinon Ashkenazy

Racah Institute of Physics, The Hebrew University of Jerusalem

I. Popov (HUJI) · W. L. Millar, V. M. Bjelland, W. Wuensch (CERN)

Montreux, Switzerland · Monday 5 October 2026



2026-10-04

Does conditioning leave a structural memory?

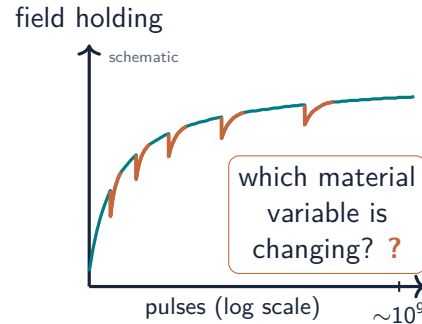
Sequel to last year's MeVArc talk. 2025 showed the dislocation model and cross-sectional STEM finding a modified near-surface layer; I closed asking for more comparative forensics—other materials, more samples. This year answers that: a large-area copper measurement testing the model's central prediction, and a stainless-steel experiment testing whether it generalizes. E_S is not on any slide (Q&A only).

Conditioning works. What does it change?



- ▶ Conditioning is how every high-gradient device reaches its usable field.^a
- ▶ Breakdown rate has an extreme field dependence, $\sim E^{30}$.^a
- ▶ It tracks the number of **high-field pulses**, not the number of breakdowns.^{a,b}
- ▶ The field itself drives conditioning, not the arcs.^{a,b}

Something in the metal stores the improvement. We do not know what.^a



Sources: ^aWuensch et al., Rev. Mod. Phys. 98, 025004 (2026); ^bDegiovanni et al., PRAB 19, 032001 (2016).

2026-10-04

Does conditioning leave a structural memory?

└ Conditioning works. What does it change?

Two macroscopic clues constrain the answer: the near-thirtieth-power field dependence, and conditioning following pulse count not breakdown count. Both say the applied field, not the arc, drives the work. Look for a field-driven change in the metal itself. (Optional aside: Walter covered this in his invited talk this morning.)

Conditioning works. What does it change?

- ▶ Conditioning is how every high-gradient device reaches its usable field.^a
- ▶ Breakdown rate has an extreme field dependence, $\sim E^{30}$.^a
- ▶ It tracks the number of **high-field pulses**, not the number of breakdowns.^{a,b}
- ▶ The field itself drives conditioning, not the arcs.^{a,b}

Something in the metal stores the improvement. We do not know what.^a

field holding

which material variable is changing? ?

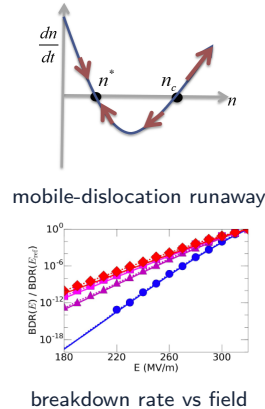
pulses (log scale) $\sim 10^9$

Sources: ^aWuensch et al., Rev. Mod. Phys. 98, 025004 (2026); ^bDegiovanni et al., PRAB 19, 032001 (2016).

Our model: breakdown as collective dislocation dynamics



- ▶ Field holding rises FCC→BCC→HCP^d — the order of barriers to dislocation motion.
- ▶ The **MDDF** model: breakdown nucleates when the mobile-dislocation population, driven by Maxwell stress, runs away.^{a,b}
- ▶ One model gets the $\sim E^{30}$ rate and its temperature dependence,^d and was shown to reproduce the histogram of dark-current spikes.^c



Central prediction: conditioning *evolves the near-surface dislocation structure*.^d

Sources: ^aEngelberg et al., PRL 120, 124801 (2018); ^bPRAB 22, 083501 (2019); ^cPRAB 23, 123501 (2020); ^dWuensch et al., RMP 98, 025004 (2026).

2026-10-04

Does conditioning leave a structural memory?

└ Our model: breakdown as collective dislocation dynamics

MDDF model is our own (2018–2020). Cartoon shows birth-death dynamics of the mobile-dislocation population: below critical density it decays, above it a runaway triggers nucleation. The plot reproduces the field dependence of breakdown rate. Key line for this talk: conditioning should leave an evolving dislocation structure just below the surface. Next slide: what that sentence means for a measurement.

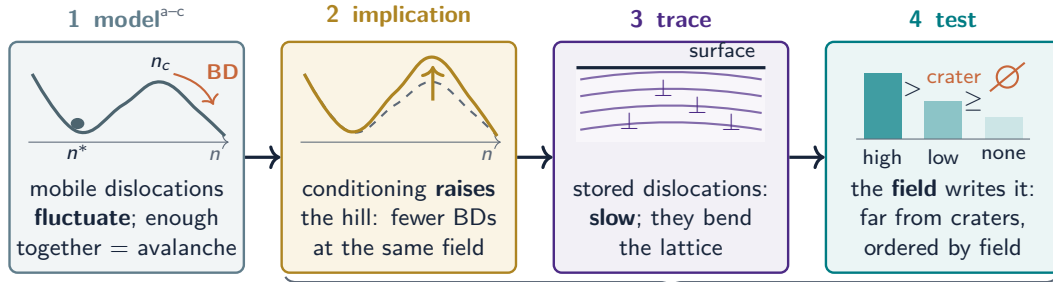
Our model: breakdown as collective dislocation dynamics

- ▶ Field holding rises FCC→BCC→HCP^a — the order of barriers to dislocation motion.
- ▶ The **MDDF** model: breakdown nucleates when the mobile-dislocation population, driven by Maxwell stress, runs away.^b
- ▶ One model gets the $\sim E^{30}$ rate and its temperature dependence,^d and was shown to reproduce the histogram of dark-current spikes.^c

Central prediction: conditioning evolves the near-surface dislocation structure.^d

Sources: ^aEngelberg et al., PRL 120, 124801 (2018); ^bPRAB 22, 083501 (2019); ^cPRAB 23, 123501 (2020); ^dWuensch et al., RMP 98, 025004 (2026).

What the model says the microscope should see



inference from the model: no equation for conditioning yet

- In the model, a $\sim 10^9$ -pulse memory must sit in stored, EBSD-visible dislocations.
- The field writes it: look far from craters; expect the order to follow the local field.

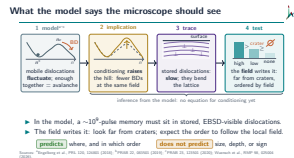
predicts where, and in which order **does not predict** size, depth, or sign

Sources: ^aEngelberg et al., PRL 120, 124801 (2018); ^bPRAB 22, 083501 (2019); ^cPRAB 23, 123501 (2020); Wuensch et al., RMP 98, 025004 (2026).

2026-10-04

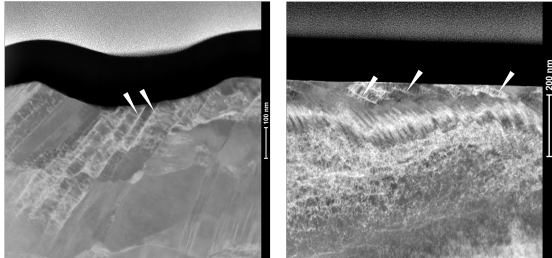
Does conditioning leave a structural memory?

What the model says the microscope should see



Slide three was one breakdown; this is what it implies for conditioning. In the model the mobile population is fast: it fluctuates and decays back. Conditioning means fewer breakdowns at the same field, so the hill must rise, and a memory lasting a billion pulses must sit in the stored dislocations that set the hill. Those bend the lattice, which EBSD reads. The field writes it, so look away from craters. The model fixes where and in which order, not size, depth, or sign.

STEM already saw it, at specific locations



Cross-sectional STEM of conditioned hard (as-machined) Cu: dislocation-poor zone in the top ~ 200 nm.^{a,b}

- ▶ **Direct** evidence of near-surface plasticity in conditioned regions.^a
- ▶ Fewer dislocation walls at 300 K and 30 K; denuded layer clearest at 30 K.^a
- ▶ Fits the clue that soft Cu conditions far more slowly than hard Cu: the starting population matters.^c
- ▶ FIB preparation plants its own dislocations; trust only large-scale structure.^b
- ▶ One cross-section cannot speak for the whole electrode.

Seen at specific locations. The large-area view was missing.

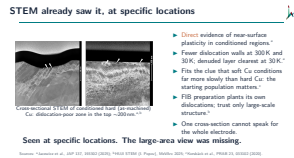
Sources: ^aJacewicz et al., JAP 137, 193302 (2025); ^bHUJI STEM (I. Popov), MeVArc 2025; ^cKorsbäck et al., PRAB 23, 033102 (2020).

2026-10-04

Does conditioning leave a structural memory?

└─ STEM already saw it, at specific locations

2025 STEM images. Dark band is the protective coating; just beneath it the field-exposed region is nearly free of dislocation walls while the reference keeps walls continuous to the surface. Published wording is conservative: 300 K shows significant reduction of walls; the denuded zone is clearly visible at 30 K. The paper gives no thickness; 200 nanometers is read from images, the number the manuscript uses. Hard-versus-soft copper contrast (Korsbäck 2020) is an independent macroscopic clue: hard copper conditioned within about 25 million pulses; soft samples began saturating near 600 million or stayed unsaturated longer, consistent with different starting dislocation populations. Two lessons: FIB preparation generates dislocations and loops, so only larger-scale structure is trustworthy; TEM and STEM are intrinsically local. Last year I asked for more comparative forensics: other materials, more samples. That is exactly the gap next slides fill.



A sub-yield stress, repeated a billion times

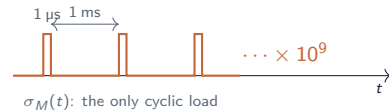


Maxwell stress

$$\sigma_M = \frac{\epsilon_0 E^2}{2}$$

at 80 MV/m: 0.028 MPa^c

~10³ × below the yield of annealed Cu^c



**The model predicts a large-area signature.
Last year we asked for it; this year we looked.**

- ▶ Fatigue at 10⁸–10¹⁰ cycles rearranges dislocations irreversibly, well below yield.^{a,b}
- ▶ Here: ~10⁹ conditioning pulses.^c
- ▶ Pulsed DC means **no RF pulsed heating** and no bulk currents. Maxwell stress is the only cyclic load.^c
- ▶ MDDF needs only the stress that moves **mobile dislocations in a frustrated crystal**: far below yield.

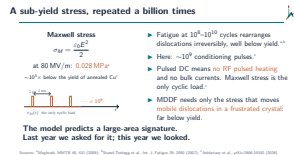
Sources: ^aMughrabi, MMTB 40, 431 (2009); ^bStanzl-Tschegg et al., Int. J. Fatigue 29, 2050 (2007); ^cAshkenazy et al., arXiv:2606.19192 (2026).

2026-10-04

Does conditioning leave a structural memory?

└ A sub-yield stress, repeated a billion times

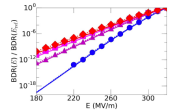
Maxwell stress is three orders of magnitude below yield. The point is not amplitude but accumulation. VHCF shows oscillatory loading well below yield still rearranges dislocations irreversibly for a billion cycles—our regime. Pulsed-DC is cleaner than RF structures: no RF magnetic field, no pulsed surface heating, no bulk Ohmic current. Maxwell stress is the only cyclic mechanical load. Earlier RF-based studies could never fully separate thermal fatigue from field effects. MDDF has no threshold at the yield stress; the relevant stress is what moves mobile dislocations in a frustrated crystal, far below yield. That argument bridges the 2025 model-and-microscopy story to this year's large-area measurement. (Optional aside: Jianyu, with Guodong Meng, and Veronika show field-driven deformation at the nanoscale this afternoon, 16:00 and 16:30.)



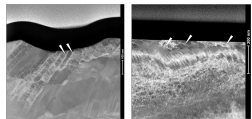
Last year we asked for more forensics. Here it is.



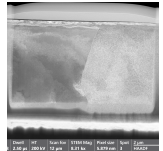
MeVArc 2025



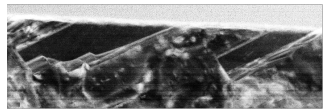
MDDF model



local STEM denuded zone^{a,b}

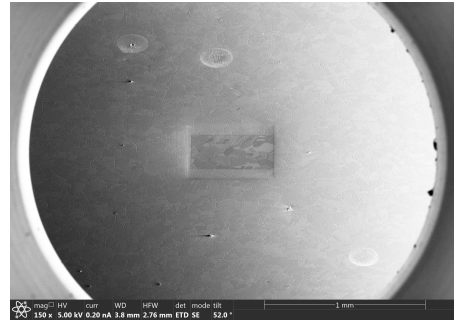


FIB lamella, HAADF overview^a



cross-section STEM at the surface^a

MeVArc 2026



Large-area EBSD across one cathode

Closed with: other materials? more samples?

Sources: ^aHUJI STEM (I. Popov), MeVArc 2025; ^bJacewicz et al., JAP 137, 193302 (2025).

2026-10-04

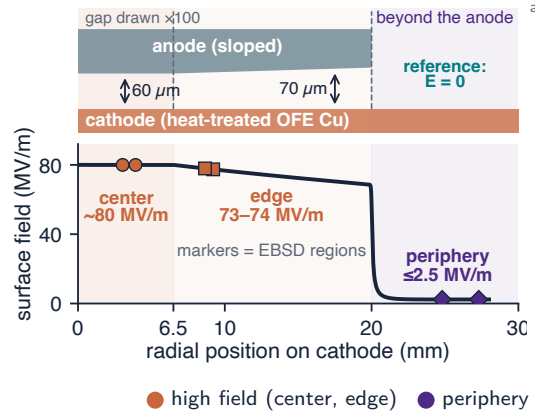
Does conditioning leave a structural memory?

└ Last year we asked for more forensics. Here it is.

Light visual bridge: 2025 showed the model and local cross-section; 2026 delivers the large-area structural comparison on one electrode. Do not quote the 75 percent effect here.



One cathode turns field exposure into a controlled coordinate



- ▶ Heat-treated OFE Cu, pulsed DC
- ▶ 1 μ s pulses at 1 kHz, $\sim 10^9$ pulses
- ▶ Peak field ~ 80 MV/m at center
- ▶ Nine EBSD regions: center, edge, periphery, and external reference

The field gradient gives us internal controls on a single specimen.

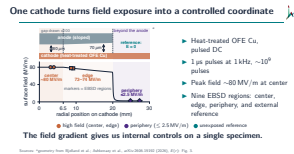
Sources: ^ageometry from Bjelland et al.; Ashkenazy et al., arXiv:2606.19192 (2026), $E(r)$: Fig. 3.

2026-10-04

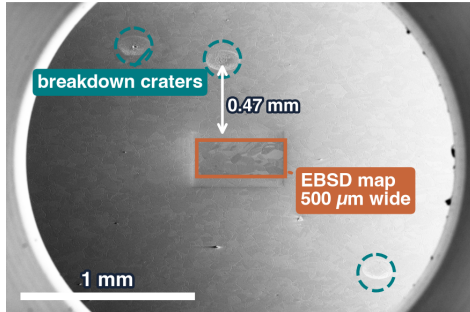
Does conditioning leave a structural memory?

One cathode turns field exposure into a controlled coordinate

One figure: cross-section on top, surface field below, same radius axis. The sloped anode maps radius onto local field: about 80 MV/m under the flat center, 73 to 74 at the edge regions under the sloped gap, below about 2.5 MV/m beyond the anode radius at 20 mm. Markers are the EBSD regions; the colors—copper for high field, purple for the low-field periphery, teal for the unexposed reference—stay the same on every data slide. Comparing positions on one cathode removes cathode-to-cathode preparation differences; the identically treated, never-installed cathode is the external reference. That geometry is the "more comparative forensics" design from last year's outlook. (Optional aside: this is CERN's vacuum breakdown lab; Victoria gave its update this morning.)



EBSD maps plastic activity far from breakdown craters



- ▶ Large area where TEM could only sample a point
- ▶ 500 μm -wide maps; 3 μm step (lateral only)^a; information depth ~ 40 nm at 15 kV^{b,c}
- ▶ $\sim 24,000$ indexed points per ROI^a
- ▶ **LAM** = local average misorientation: lattice tilt between a point and its neighbors (KAM, LOS: cross-checks)^a
- ▶ Misorientation is a proxy for GND density^d

We measure the cumulative response to the field, not crater damage.^a

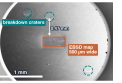
Sources: ^aAshkenazy et al., arXiv:2606.19192 (2026); ^b ~ 40 nm at 15 kV: CASINO Monte Carlo, Drouin et al., Scanning 29, 92 (2007); ^cmeasured 38–72 nm over 5–30 kV, Chen et al., Ultramicroscopy 111, 1488 (2011); ^dPantleon, Scr. Mater. 58, 994 (2008).

2026-10-04

Does conditioning leave a structural memory?

└ EBSD maps plastic activity far from breakdown craters

EBSD maps plastic activity far from breakdown craters



- ▶ Large area where TEM could only sample a point
- ▶ 500 μm -wide maps; 3 μm step (lateral only)^a; information depth ~ 40 nm at 15 kV^{b,c}
- ▶ $\sim 24,000$ indexed points per ROI^a
- ▶ **LAM** = local average misorientation: lattice tilt between a point and its neighbors (KAM, LOS: cross-checks)^a
- ▶ Misorientation is a proxy for GND density^d

We measure the cumulative response to the field, not crater damage.^a

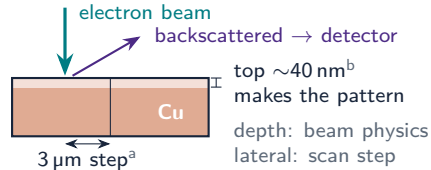
Sources: ^aAshkenazy et al., arXiv:2606.19192 (2026); ^b ~ 40 nm at 15 kV: CASINO Monte Carlo, Drouin et al., Scanning 29, 92 (2007); ^cmeasured 38–72 nm over 5–30 kV, Chen et al., Ultramicroscopy 111, 1488 (2011); ^dPantleon, Scr. Mater. 58, 994 (2008).

EBSD chosen for large-area where TEM is local. On the SEM overview: the copper box is one 500-micron EBSD map after a shallow FIB clean that removes the oxide (under 10 nm by AFM); the dashed teal circles are breakdown craters, the nearest about half a millimeter away. The 3 micron step and kernel fix only lateral resolution; backscattered electrons come from the top 40 nanometers of copper at 15 kV, so this is a near-surface measurement. All ROIs deliberately far from craters, measuring the cumulative material response to the field, not arc damage. LAM, local average misorientation, is the primary metric; KAM and LOS check the result does not depend on one analysis choice.

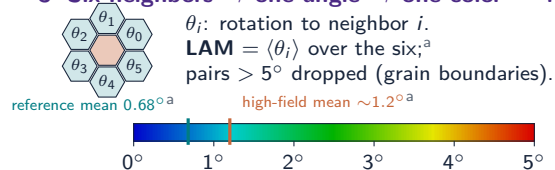
What one EBSD pixel measures



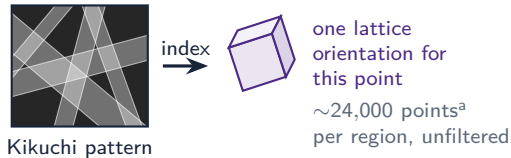
1 15 kV electron beam, point by point



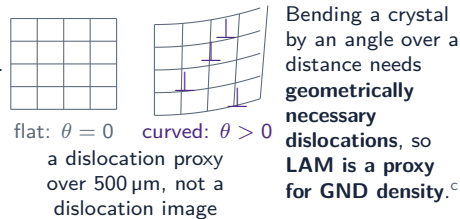
3 Six neighbors → one angle → one color



2 Pattern → lattice orientation



4 Why a rotation angle counts dislocations



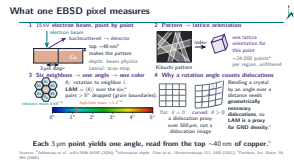
Each 3 μm point yields one angle, read from the top ~40 nm of copper.^b

Sources: ^aAshkenazy et al., arXiv:2606.19192 (2026); ^binformation depth: Chen et al., Ultramicroscopy 111, 1488 (2011); ^cPantleon, Scr. Mater. 58, 994 (2008).

2026-10-04

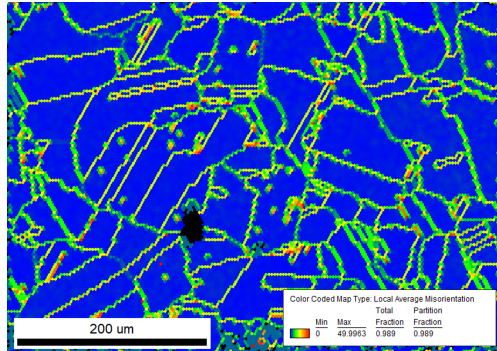
Does conditioning leave a structural memory?

What one EBSD pixel measures

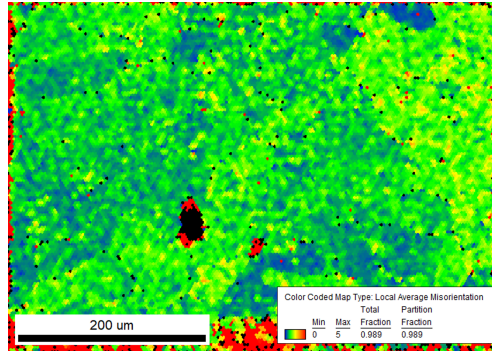


Every colored pixel on the next slides is this one angle. The beam reads the top forty nanometers of copper at each point of a three-micron grid; the Kikuchi pattern gives one lattice orientation per point. LAM averages the rotation to the six neighbors, dropping pairs above five degrees, so grain boundaries are excluded. A bent lattice needs geometrically necessary dislocations, so LAM is a dislocation proxy over half a millimeter, not a dislocation image.

How to read a LAM map: wide-angle vs low-angle



0–50°: grain boundaries dominate



0–5° scale: intragrain curvature

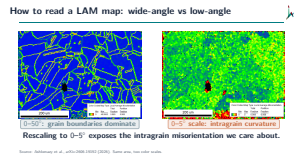
Rescaling to 0–5° exposes the intragrain misorientation we care about.

Source: Ashkenazy et al., arXiv:2606.19192 (2026). Same area, two color scales.

2026-10-04

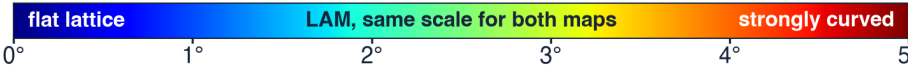
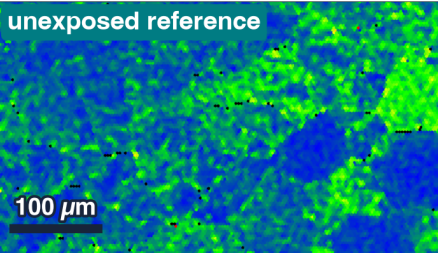
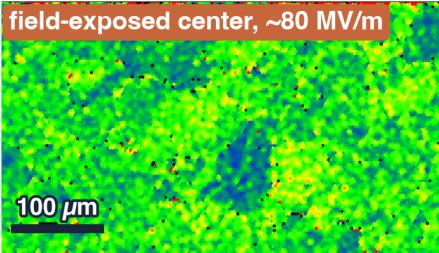
Does conditioning leave a structural memory?

How to read a LAM map: wide-angle vs low-angle



Teach the metric before comparison. At wide 0-50 degree scale only grain boundaries are visible. Rescaling to 0-5 degrees reveals intragrain lattice curvature—the low-angle misorientation proxying geometrically necessary dislocations. Every LAM map following uses this same 0-5 degree scale.

High-field copper contains more intragrain curvature



Same scale, same magnification: field-exposed grains carry more curvature.

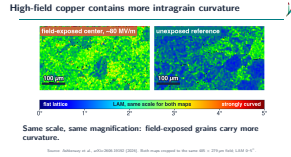
Source: Ashkenazy et al., arXiv:2606.19192 (2026). Both maps cropped to the same $485 \times 279 \mu\text{m}$ field; LAM $0\text{--}5^\circ$.

2026-10-04

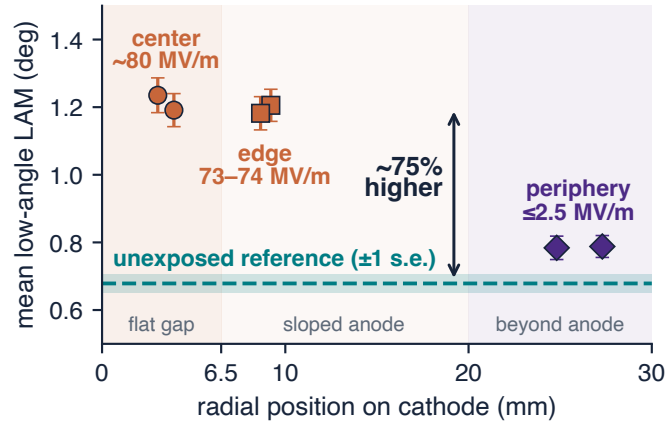
Does conditioning leave a structural memory?

High-field copper contains more intragrain curvature

Same color scale and now the same magnification, both maps cropped to the same field of view with identical hundred-micron bars. Look at the spatial texture, not just the average color: the field-exposed region carries far more green and yellow, high-misorientation structure inside the grains, where the reference stays blue. Pair shown is representative; the quantitative conclusion uses all nine ROIs.



The signal separates into three exposure tiers



- ▶ Center and edge: $\sim 1.2^\circ$
- ▶ External reference: $\sim 0.68^\circ$
- ▶ High field is $\sim 75\%$ higher
- ▶ Grain size does not follow this hierarchy

The ordering follows field history, not sample position or grain size.

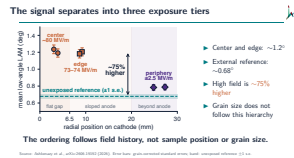
Source: Ashkenazy et al., arXiv:2606.19192 (2026). Error bars: grain-corrected standard errors; band: unexposed reference ± 1 s.e.

2026-10-04

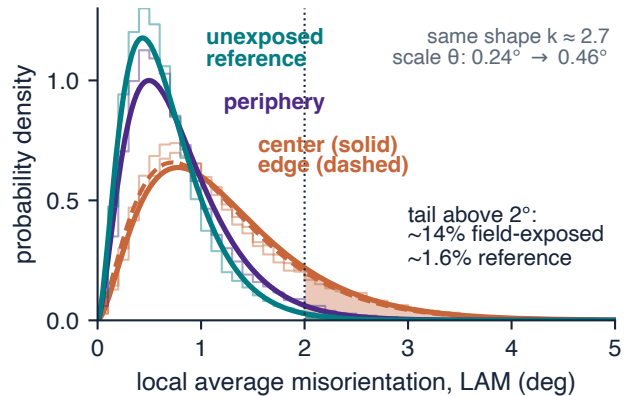
Does conditioning leave a structural memory?

The signal separates into three exposure tiers

Same colors as the geometry slide: copper is high field, purple the periphery, teal the reference band. Three tiers: high-field center and edge highest and similar; field-exposed periphery intermediate; unexposed references lowest. Periphery is powerful: it shared machining, heat treatment, mounting, vacuum, and handling with high-field regions but saw at most 2.5 MV/m—a low-field control, not zero-field. Intermediate value consistent with residual fringe-field exposure. The 75 percent figure characterizes this cathode and history, not a universal constant.



Field exposure builds a high-misorientation tail



LAM tail above 2°
~0.14 field-exposed
~0.016 reference
~8× increase

Gamma shape $k \approx 2.7$ fixed; the scale roughly doubles.

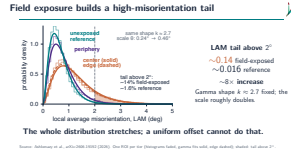
The whole distribution stretches; a uniform offset cannot do that.

Source: Ashkenazy et al., arXiv:2606.19192 (2026). One ROI per tier (histograms faded, gamma fits solid, edge dashed); shaded: tail above 2°.

2026-10-04

Does conditioning leave a structural memory?

Field exposure builds a high-misorientation tail



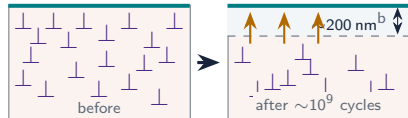
Tier colors again; the dotted line is two degrees and the shaded areas are the tails beyond it. Low-angle distribution broadens and grows a long tail: fraction above two degrees rises roughly eightfold. Gamma shape parameter stays near 2.7 while scale roughly doubles—a fixed-shape stretch, signature of progressive deformation. LAM, KAM, and LOS agree. KS, chi-squared, and Anderson-Darling separate the distributions; every center or edge ROI differs from unexposed ones with KS distance above 0.25. Quote effect sizes and grain-corrected errors rather than extreme pixel-pair p-values.

The large-area test supports the model's prediction



Hard, as-machined Cu

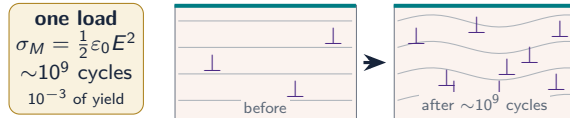
dense starting population (2025 STEM cathodes)



dislocations escape to the surface or annihilate $\Rightarrow$ **depleted layer** (STEM)^b

Heat-treated OFE Cu

sparse starting population (this cathode)



the manuscript's interpretation^a same load generates and rearranges dislocations $\Rightarrow$ **lattice curvature** (EBSD, LAM $\sim 75\%$ higher)^a

- To our knowledge, the first **large-area** observation that conditioned regions carry a different near-surface dislocation structure, ordered as MDDF predicts.^a
- The measured order follows the simulated local field:
center $\approx$ edge $>$ periphery $>$ reference.^a

Before, dislocations were plausible.

Now, a dislocation-sensitive state maps the conditioning history.

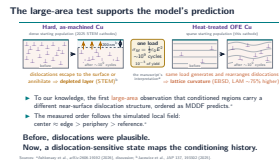
Sources: ^aAshkenazy et al., arXiv:2606.19192 (2026), discussion; ^bJacewicz et al., JAP 137, 193302 (2025).

2026-10-04

Does conditioning leave a structural memory?

└ The large-area test supports the model's prediction

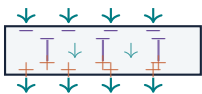
The measured hierarchy is the large-area structural signature the model predicted and STEM saw locally. The cartoon is the manuscript's interpretation of why the surface denuded zone and the EBSD curvature are not in conflict: one mechanism acting on two very different starting populations. Left, dense as-machined copper: cyclic stress lets near-surface dislocations escape or annihilate—the depleted layer STEM saw, about 200 nanometers. Right, heat-treated copper measured here: little to deplete, so the same load generates and rearranges dislocations—the curvature EBSD records. Size, depth and sign were not predicted; the sign is set by the starting population. Next slide gives the length scale; the depth connection is still open. If asked about E_S (not on the slides): the spatial ordering matches the simulated E_S profile, so the dislocation population is a candidate structural basis for it. Never say we measured E_S .



Elastic screening: dislocations respond like bound charge



Dielectric in an applied field

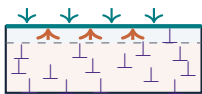


applied field E_0
bound charge polarizes
interior field
reduced to E_0/ϵ_r

1 $\lesssim 0.1 \text{ nm}$
Field penetration
Thomas–Fermi:
the field stops at
the surface.

2 $\ell_D \propto 1/\sqrt{\rho} \approx 25 \text{ nm}^a$ (order of magnitude)
Debye-like screening of fluctuations
At $\rho \sim 10^{14} \text{ m}^{-2}$, only dislocations within
 $\sim \ell_D$ of the surface exchange fluctuating stress
with it; deeper ones are driven by the load but
silent at the surface. Observable: LAM; denuded
 $\sim 200 \text{ nm} \approx 8 \ell_D$.^d

Copper under the Maxwell traction



1 field stops at the surface
2 σ_M enters as stress
3 σ_M enters as stress
dislocations as bound
charge (dipole regime)^{b,c}

3 $\sim \text{mm}$
Applied load: not screened
Dislocation stress is
self-equilibrated; ordinary
elasticity over the loading
geometry.^d

A plasma of dipoles: Debye-like for fluctuations, dielectric-like for the mean load; it reduces, it does not nullify.
The surface is an absorbing wall: a sheath of $\sim \ell_D$ sets what the surface sees, hence emission and breakdown.

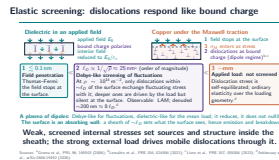
Weak, screened internal stresses set sources and structure inside the sheath; the strong external load drives mobile dislocations through it.

Sources: ^aGroma et al., PRL 96, 165503 (2006); ^bLemaître et al., PRE 104, 024904 (2021); ^cLivne et al., PRE 107, 055004 (2023); ^dAshkenazy et al., arXiv:2606.19192 (2026).

2026-10-04

Does conditioning leave a structural memory?

└ Elastic screening: dislocations respond like bound charge



Three scales: the field penetrates below 0.1 nanometer (Thomas–Fermi); fluctuations are screened within about 25 nanometers; the mean load is not screened and spreads over millimeters. Screening is Debye–Hückel-like: a 2D two-component Coulomb gas of positive and negative dislocations with the flow stress as temperature; $\ell_D \propto 1/\sqrt{\rho}$ is an order-of-magnitude estimate (2D single slip, Groma 2006); the picture does not depend on its details. The mean stress cannot be screened: dislocation fields are self-equilibrated; the free surface is a sink, so no pile-up back stress. Q&A parallels: plasma sheath at an absorbing wall, only the sheath population sets the wall flux; semiconductor depletion layer, many Debye lengths wide when surface potential far exceeds kT; dielectric, reduces but does not nullify. Self-consistency: tenfold depletion triples ℓ_D . Depletion-width scaling is analogy, not derivation.

Copper establishes the effect, not universality



What is established

- ▶ Large-area structural difference
- ▶ Three independent metrics
- ▶ Internal and external controls
- ▶ Supports the MDDF prediction

What remains open

- ▶ One conditioned cathode, one experiment
- ▶ Full depth profile
- ▶ Grain-orientation dependence
- ▶ More conditioned cathodes
- ▶ Cross-material generality

Limits of this dataset: field level and breakdown density vary together by construction ($\sim 24, 12, 5$ breakdowns per cm^2 , center outward); LAM values are relative, compared only between regions mapped under the same step and protocol.

A stronger test changes both the material and the loading history.

Source: Ashkenazy et al., arXiv:2606.19192 (2026), discussion and limitations.

2026-10-04

Does conditioning leave a structural memory?

└ Copper establishes the effect, not universality

The honest limitations: one conditioned cathode and one external reference; field level and breakdown density covary; LAM is a relative proxy. The invitation to replicate and compare across facilities comes on the last slide. Depth matters: STEM depletion in hard copper, EBSD curvature in heat-treated copper, each in the top few hundred nanometers at most. A depth profile on one material would show whether they are two ends of one redistribution. Next experiments add depth and change material and loading mode.

Copper establishes the effect, not universality

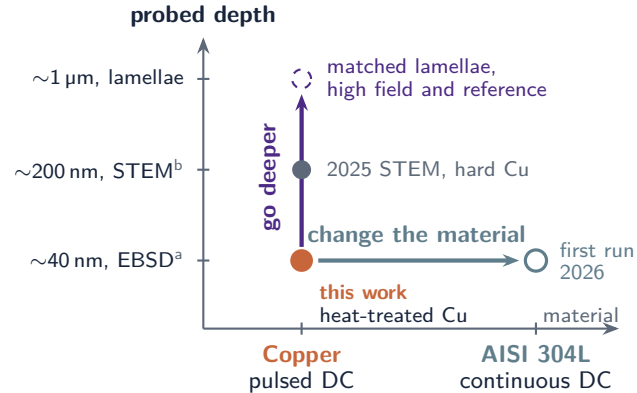
What is established	What remains open
▶ Large-area structural difference ▶ Three independent metrics ▶ Internal and external controls ▶ Supports the MDDF prediction	▶ One conditioned cathode, one experiment ▶ Full depth profile ▶ Grain-orientation dependence ▶ More conditioned cathodes ▶ Cross-material generality

Limits of this dataset: field level and breakdown density vary together by construction ($\sim 24, 12, 5$ breakdowns per cm^2 , center outward); LAM values are relative, compared only between regions mapped under the same step and protocol.

A stronger test changes both the material and the loading history.

Source: Ashkenazy et al., arXiv:2606.19192 (2026), discussion and limitations.

Next: go deeper, and change the material



- ▶ Matched high-field and reference FIB lamellae; TEM, STEM and EBSD through the top $\sim 1\ \mu\text{m}$
- ▶ Artifact controls from the 2025 “red herrings”
- ▶ One depth profile from STEM depletion to EBSD curvature?
- ▶ Stainless steel under continuous DC, with Consorzio RFX — next slide

Depth continues the Uppsala STEM line; material tests the mechanism itself.

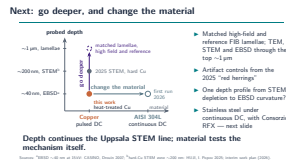
Sources: ^aEBSD $\sim 40\ \text{nm}$ at 15 kV: CASINO, Drouin 2007; ^bhard-Cu STEM zone $\sim 200\ \text{nm}$: HUJI, I. Popov 2025; interim work plan (2026).

2026-10-04

Does conditioning leave a structural memory?

Next: go deeper, and change the material

Two coordinates on one picture. Today’s result sits at about 40 nanometers, EBSD in heat-treated copper; the 2025 STEM depletion was about 200 nanometers deep, in hard copper. Going deeper means matched high-field and reference FIB lamellae, TEM, STEM and EBSD through the top micron, with the artifact controls learned in 2025, aiming at one depth profile from STEM depletion to EBSD curvature. Along the material axis: AISI 304L under continuous DC, a first campaign with Consorzio RFX. Next slide.



Steel: different material, different structural evolution



What differs from copper

- ▶ Lower stacking-fault energy: planar slip, different dislocation mobility
- ▶ Continuous DC: a steady traction of $\sim 16 \text{ kPa}^a$, not $\sim 10^9$ pulses
- ▶ Metastable austenite is a built-in strain gauge: ϵ and α' martensite mark plastic strain above a few $0.1\% ^b$

The dislocation picture expects a different near-surface evolution. The test: does any field-correlated change appear at all?



Consorzio RFX — De Lorenzi, Pilan, Spada

RFX HV program: N. Pilan, next talk (12:30).

Copper supports the memory; steel asks whether it is universal.

Sources: ^a $\sim 16 \text{ kPa} = \epsilon_0 E^2 / 2$ at 60 MV/m ; ^bBayerlein et al., MSEA 114, L11 (1989); ^cRFX project record, EBSD pass of 29 Sep 2026.

First run with Consorzio RFX^c

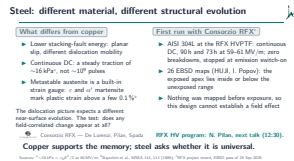
- ▶ AISI 304L at the RFX HVPTF: continuous DC, 90 h and 73 h at 59–61 MV/m; zero breakdowns, stopped at emission switch-on
- ▶ 26 EBSD maps (HUJI, I. Popov): the exposed apex lies inside or below the unexposed range
- ▶ Nothing was mapped before exposure, so this design cannot establish a field effect

2026-10-04

Does conditioning leave a structural memory?

└ Steel: different material, different structural evolution

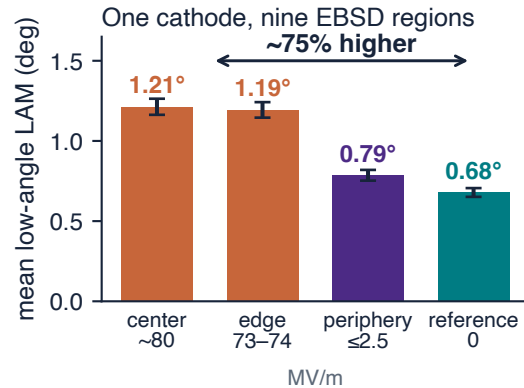
Steel is a different material, so the dislocation picture expects a different structural evolution: lower stacking-fault energy and planar slip, and a steady traction of about 16 kilopascals instead of about a billion pulses. Its metastable austenite is a built-in strain gauge: epsilon and alpha-prime martensite appear above a few tenths of a percent of plastic strain. With Consorzio RFX—Antonio De Lorenzi, Nicola Pilan and Emanuele Spada—two 304L electrodes ran 90 and 73 hours at 59 to 61 MV/m, zero breakdowns. The exposed apex lies inside or below the unexposed range; nothing was mapped before exposure, so no field effect can be established. For Q&A, what next on steel: one electrode mapped before and after being driven into breakdowns, with epsilon and alpha-prime in the phase list so that absent martensite becomes a strain bound. Also for Q&A: we have not yet looked for martensite on an exposed surface; the plan is to use its absence as a strain gauge. The copper signal itself is a low-strain rearrangement (a 0.5 deg LAM rise over a $3 \mu\text{m}$ step is of order $10^{13} \text{ m}^{-2} \text{ GND}$), so steel and copper would then say the same thing from two sides. Nicola's talk follows.



Conditioning appears to leave a subsurface memory



1. Our model predicts conditioning evolves the near-surface dislocation structure.
2. 2025 STEM saw it locally; **large-area EBSD now supports it** (~75% higher misorientation).
3. Depth and stainless steel test whether the memory survives a change of scale, alloy, and loading.



Last year: “other materials? more samples?” — here is the large-area answer.

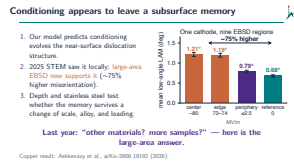
Copper result: Ashkenazy et al., arXiv:2606.19192 (2026).

2026-10-04

Does conditioning leave a structural memory?

Conditioning appears to leave a subsurface memory

Close the loop from the title and from last year. Model predicted a structural memory; STEM saw it at a few specific locations; large-area EBSD supports it across the electrode: the four bars are the tier means, 1.21 and 1.19 degrees at center and edge, 0.79 at the periphery, 0.68 on the reference, about 75 percent higher at high field. Stainless-steel and depth programs are the next round of comparative forensics promised in 2025. The invitation and the paper link are on the next slide. Cite arXiv:2606.19192.



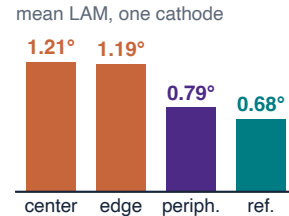
Conclusions, and an invitation



1. The model predicted a structural memory of conditioning.
2. STEM saw it locally; large-area EBSD now supports it.
3. Open: depth^a, replication, other materials — a stainless-steel test with Consorzio RFX is under way.^b

Talk to me: new materials, other electrode configurations, cross-facility comparisons, same-area before-and-after mapping — let us test this together.

Sources: ^aDepth program: interim work plan (2026); ^bstainless-steel test: RFX project record (29 Sep 2026).



arxiv.org/pdf/2606.19192

Ashkenazy et al., arXiv:2606.19192

2026-10-04

Does conditioning leave a structural memory?

Conclusions, and an invitation

Three lines: the model predicted a structural memory, STEM saw it locally, and large-area EBSD now supports it. Open: depth, replication on more cathodes, other materials—and the stainless-steel test with Consorzio RFX is under way. Then the invitation: new materials, other electrode configurations, cross-facility comparisons, same-area before-and-after mapping. The QR code and link open the paper. (Optional aside: Mircea's resistivity measurements, just before this talk, probe the same state.) Thank you.

Conclusions, and an invitation

1. The model predicted a structural memory of conditioning.
2. STEM saw it locally; large-area EBSD now supports it.
3. Open: depth^a, replication, other materials — a stainless-steel test with Consorzio RFX is under way.^b

Talk to me: new materials, other electrode configurations, cross-facility comparisons, same-area before-and-after mapping — let us test this together.

Sources: ^aDepth program: interim work plan (2026); ^bstainless-steel test: RFX project record (29 Sep 2026).

Backup: RFX first look (preliminary)

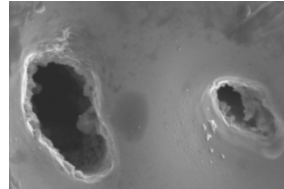


Preliminary — no field effect claimed

- ▶ AISI 304L at RFX HVPTF; continuous DC: 90 h (#3-25), 73 h (#4-25); peak 59–61 MV/m; zero breakdowns (stopped at emission switch-on)
- ▶ SEM: “exploded” near-surface inclusions (3–7 μm) on field-exposed samples; reference cavities round and clean^a
- ▶ EBSD: #1-25, #2-25, #3-25 index (97–99.5%); #4-25: two maps, one an outlier on confidence index, position unrecorded
- ▶ Apex-vs-side contrast reproduces on never-installed #2-25: forming geography



#1-25 reference: round, clean (HFW 16.6 μm)^a



#3-25 field-exposed: “exploded” (HFW 10 μm)^a



Sectioned electrode

Selected panels; the full SEM set is in the RFX SEM report.

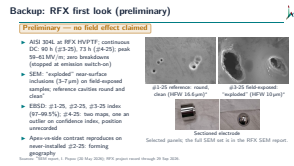
Sources: ^aSEM report, I. Popov (20 May 2026); RFX project record through 29 Sep 2026.

2026-10-04

Does conditioning leave a structural memory?

└ Backup: RFX first look (preliminary)

Backup for questions on the stainless-steel first look. SEM survey plus EBSD quality: three samples index well; the field-exposed #4-25 has two maps, one of them an outlier on confidence index, and its position on the electrode was not recorded, so the cause is open. The July apex-versus-side contrast reproduced on a never-installed reference, so it reflects forming geography. No structural field effect is claimed.



Backup: RFX sample and data status (29 Sep 2026)



Sample	Role	Exposure	Status
#1-25	reference	none	SEM; EBSD apex + side, 97% indexed
#2-25	reference	none	EBSD apex (18 Aug, 3 maps) + side; LAM done
#3-25	field-exposed	90 h DC, peak 61 MV/m	SEM; EBSD apex + side, 99.5% indexed; LAM done
#4-25	field-exposed	73 h DC, peak 59 MV/m	SEM/EDS; EBSD two maps, one an outlier on confidence index

- ▶ Received since July: conditioning report and $E(\theta)$; alloy confirmed as 304L, mill certificate unrecoverable; electropolishing partly documented; all 26 catalog EBSD maps (36 scans), 28 Sep.
- ▶ XRD: δ -ferrite content differs between the reference and exposed pairs, so the clean comparison is apex vs side within one electrode.
- ▶ Wide-vs-narrow map class = grain size and indexing quality (also on never-installed #2-25), not field.
- ▶ Still open: thermal history of #3-25 and #4-25 (vacuum firing?); matrix Cr/Ni EDS on all four under identical conditions; SEM θ -survey as the first exposure marker.

2026-10-04

Does conditioning leave a structural memory?

Backup: RFX sample and data status (29 Sep 2026)

Use only if asked about readiness or missing inputs. Reflects the written project record through 29 September 2026.

Sample	Role	Exposure	Status
#1-25	reference	none	SEM; EBSD apex + side, 97% indexed
#2-25	reference	none	EBSD apex (18 Aug, 3 maps) + side; LAM done
#3-25	field-exposed	90 h DC, peak 61 MV/m	SEM; EBSD apex + side, 99.5% indexed; LAM done
#4-25	field-exposed	73 h DC, peak 59 MV/m	SEM/EDS; EBSD two maps, one an outlier on confidence index

▶ Received since July: conditioning report and $E(\theta)$; alloy confirmed as 304L, mill certificate unrecoverable; electropolishing partly documented; all 26 catalog EBSD maps (36 scans), 28 Sep.

▶ XRD: δ -ferrite content differs between the reference and exposed pairs, so the clean comparison is apex vs side within one electrode.

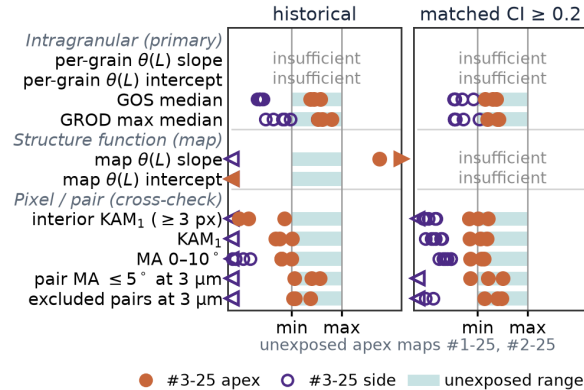
▶ Wide-vs-narrow map class = grain size and indexing quality (also on never-installed #2-25), not field.

▶ Still open: thermal history of #3-25 and #4-25 (vacuum firing?); matrix Cr/Ni EDS on all four under identical conditions; SEM θ -survey as the first exposure marker.

Backup: RFX apex vs unexposed range (29 Sep 2026)



Null so far — not a field effect, not proof of saturation



Wide map class, 2 of 4 panels (full board: RFX record). Only point above range: map $\theta(L)$ slope ($n = 2$ baseline).

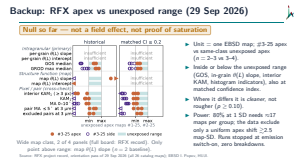
Source: RFX project record, orientation pass of 29 Sep 2026 (all 26 catalog maps); EBSD I. Popov, HUJI.

- ▶ Unit = one EBSD map; #3-25 apex vs same-class unexposed apex ($n = 2-3$ vs $3-4$).
- ▶ Inside or below the unexposed range (GOS, in-grain $\theta(L)$ slope, interior KAM, histogram indicators), also at matched confidence index.
- ▶ Where it differs it is *cleaner*, not rougher ($p \geq 0.10$).
- ▶ Power: 80% at 1 SD needs ≈ 17 maps per group; the data exclude only a uniform apex shift $\gtrsim 2.5$ map-SD. Runs stopped at emission switch-on, zero breakdowns.

2026-10-04

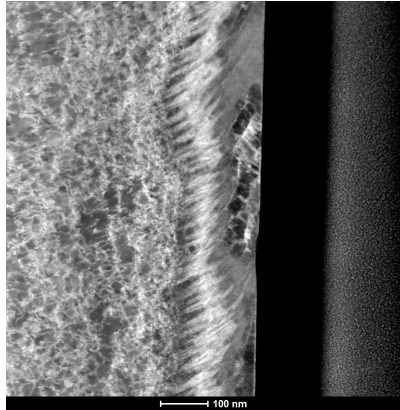
Does conditioning leave a structural memory?

Backup: RFX apex vs unexposed range (29 Sep 2026)

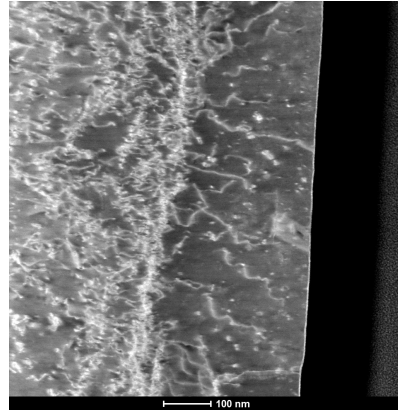


Use only if someone asks whether the steel shows anything. The exposed apex sits inside or below the unexposed apex maps of the same grain-size class, including when we match indexing confidence; the one value above is a map-level slope against only two baseline maps. That is a null at low power, not evidence of absence, and not evidence for the saturation hypothesis. The dose was about ten to the minus four of yield and the runs never reached breakdown conditioning, so the next campaign is annealed versus as-received steel, each with a sham electrode, driven into breakdowns.

Backup: field-exposed vs reference STEM (2025)



Field-exposed, 300 K: fewer walls^{a,b}



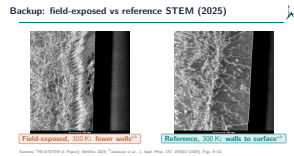
Reference, 300 K: walls to surface^{a,b}

Sources: ^aHUJI STEM (I. Popov), MeVArc 2025; ^bJacewicz et al., J. Appl. Phys. 137, 193302 (2025), Figs. 8–10.

2026-10-04

Does conditioning leave a structural memory?

└ Backup: field-exposed vs reference STEM (2025)



The field-exposed versus reference STEM pair from the 300 K cathode. In the published paper this pair shows a strong reduction of dislocation walls (Fig. 9); the zone the paper calls dislocation-denuded and clearly visible is the 30 K cathode (Fig. 10). No thickness is stated in the paper.